

PuuD-Linked Urate Oxidation to Allantoin: A Reusable Three-Reaction Bacterial Module

A commissioned review-style synthesis for a molecular-biology audience

1. Executive Summary

The stereospecific degradation of uric acid to (S)-allantoin is not, as textbooks long implied, a single-enzyme event. It is a **three-reaction relay** in which a urate oxidase generates the unstable intermediate **5-hydroxyisourate (HIU)**, a **HIU hydrolase** converts HIU to the equally unstable **2-oxo-4-hydroxy-4-carboxy-5-ureidoimidazoline (OHCU)**, and an **OHCU decarboxylase** finally produces the (S)-enantiomer of allantoin. The topic of this review — **PuuD-linked urate oxidation** — concerns a specific bacterial implementation of the *first* step, in which the oxidase is not the familiar soluble, cofactor-independent uricase (Uox) but an **integral-membrane, cytochrome c-containing protein of the DUF989/COG3748 family, designated PuuD**. This enzyme oxidizes urate by **cytochrome c-mediated electron transfer rather than by the hydrogen-peroxide-producing chemistry of soluble Uox** [PMID: 26349049]. The two downstream enzymes (PucM/UraH and UraD/PucL) are shared with the classical Uox pathway, so the "PuuD-linked" module is best understood as one of several evolutionarily interchangeable ways of feeding a conserved downstream trap-and-decarboxylate machine.

Three points define the boundaries of the system and recur throughout this review. First, the initiating oxidase is the **variable, lineage-specific node**: at least three mechanistically non-homologous enzymes — soluble Uox ($O_2 \rightarrow H_2O_2$), membrane cytochrome c PuuD (electron transfer, no H_2O_2), and the FAD-dependent HpxO — independently converge on HIU. Second, the two downstream enzymes are **ancient and conserved**, belong to well-characterized folds (a transthyretin-related β -sandwich and a novel cofactor-independent α -helical fold), and exist chiefly to **capture unstable intermediates before they decay non-enzymatically and to enforce (S)-allantoin stereochemistry** demanded by downstream allantoinase. Third, the name "PuuD" is **ambiguous**: the urate-oxidase PuuD is entirely unrelated to the well-known *E.*

coli PuuD γ -glutamyl- γ -aminobutyrate hydrolase of putrescine catabolism, and the aerobic HIU/OHCU/allantoin module must also be distinguished from a recently described anaerobic gut-bacterial route that degrades urate to xanthine and short-chain fatty acids.

This report synthesizes eight confirmed findings drawn from 18 primary papers into a coherent mechanistic and evolutionary model. The strongest conclusions — the identity and electron-transfer mechanism of PuuD, the three-enzyme architecture, the fold assignments of the two downstream enzymes, and the ancestral status of the HIU-hydrolase activity within the transthyretin family — rest on crystallography, gene-deletion/complementation, steady-state kinetics, and ancestral-sequence reconstruction. The most important open question concerns the atomic-resolution structure and electron-acceptor chemistry of PuuD itself, which remains the least-characterized enzyme in the pathway.

2. Definition and Biological Boundaries

2.1 What is included

The system comprises exactly **three sequential enzymatic activities** operating on freely diffusing (and chemically unstable) metabolites:

Step	Substrate $\rightarrow$ Product	Enzyme (this module)	Enzyme family / fold
1	Urate + O ₂ $\rightarrow$ 5-hydroxyisourate (HIU)	PuuD (DUF989/COG3748)	Integral-membrane cytochrome c; 8-TM domain
2	HIU $\rightarrow$ OHCU	HIU hydrolase (PucM/UraH)	Transthyretin-related protein (TRP), homotetramer
3	OHCU $\rightarrow$ (S)-allantoin (+ CO ₂)	OHCU decarboxylase (UraD/PucL)	Novel all- α fold, homodimer, cofactor-independent

The scope is deliberately restricted to the **PuuD-linked implementation of step 1**. The downstream steps 2 and 3 are shared with, and were originally elucidated in, the soluble-Uox version of the pathway; they are included here because they are obligatory for producing the correct product.

2.2 Neighboring processes that are frequently confused and should be treated separately

Several distinct systems share vocabulary or chemistry with this module and are commonly conflated:

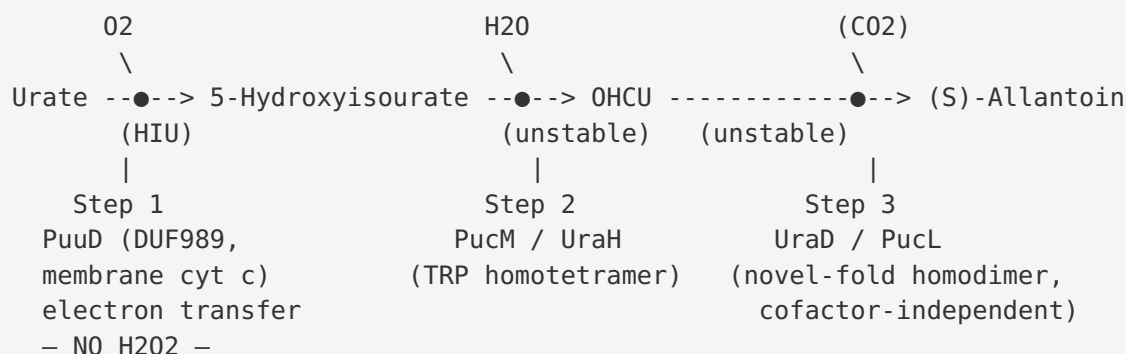
- **Soluble cofactor-independent urate oxidase (Uox).** This is the classic uricase, a cofactor-independent enzyme that reduces O_2 to **hydrogen peroxide**. It performs the *same net transformation* (urate $\rightarrow$ HIU) as PuuD but by fundamentally different chemistry. Conflating PuuD's electron-transfer mechanism with Uox's H_2O_2 chemistry is precisely the error this review's scope warns against [PMID: 26349049].
- **FAD-dependent urate hydroxylase HpxO.** In *Klebsiella pneumoniae*, urate oxidation is carried out by HpxO, an enzyme resembling flavin-dependent aromatic-ring oxygenases rather than any classical uricase [PMID: 18849434]. It is a *third*, independent solution to step 1.
- **The *E. coli* PuuD of the putrescine (Puu) pathway.** This is **γ -glutamyl- γ -aminobutyrate hydrolase**, a Cys-114 amidohydrolase with no role in purine or urate metabolism [PMID: 16499623]. It shares only a gene symbol with the urate-oxidase PuuD.
- **The anaerobic gut-bacterial urate-degradation cluster.** A widely distributed bacterial gene cluster degrades uric acid **anaerobically to xanthine or short-chain fatty acids**, never passing through HIU/OHCU/(S)-allantoin and not using PuuD-type oxidase chemistry [PMID: 37541197].
- **Downstream ureide catabolism (allantoinase, allantoin racemase, allantoinase).** These enzymes act *after* allantoin and belong to the broader purine-nitrogen assimilation network, not to the three-reaction module itself. Allantoin racemase is relevant only as an explanation of why stereochemical control at step 3 matters (see §6).

2.3 Competing definitions

The chief definitional tension in the literature is whether "the urate degradation pathway" means (a) the single oxidase step (the historical, pre-2006 view) or (b) the full three-enzyme relay. The modern, well-supported definition is (b): urate oxidation alone yields racemic allantoin over hours, whereas the complete three-enzyme complement yields dextrorotatory (S)-allantoin within seconds [PMID: 16462750]. A second tension concerns whether steps 2 and 3 are two proteins or one — in some lineages they are fused into a single bifunctional enzyme (see §5).

3. Mechanistic Overview

3.1 The best current model of the sequence of events



Step 1 (obligatory, variable enzyme). PuuD oxidizes urate to HIU. Uniquely among the three urate-oxidase solutions, PuuD is an integral-membrane protein whose 8-transmembrane-helix domain corresponds to DUF989 and whose C-terminal cytochrome c mediates electron transfer. Deletion of *puuD* in *Agrobacterium fabrum* abolishes urate degradation; complementation with a cytosolic zebrafish Uox fully restores it, demonstrating that PuuD occupies the urate-oxidase slot. Critically, the zebrafish-Uox-complemented strain shows a ~4-fold increase in catalase (a proxy for H₂O₂ production) whereas wild-type PuuD shows none — direct evidence that PuuD oxidizes urate by **cytochrome c-mediated electron transfer, not by H₂O₂ chemistry** [PMID: 26349049].

Step 2 (obligatory, conserved enzyme). HIU is chemically unstable and would decay non-enzymatically to racemic allantoin. HIU hydrolase (PucM in *Bacillus subtilis*, UraH/KpHIUH in *Klebsiella*) captures HIU and hydrolyzes it to OHCU. It is a **homotetrameric transthyretin-related protein (TRP)** with conserved active-site residues (His-14, Arg-49, His-105 and a C-terminal Tyr-Arg-Gly-Ser motif in *B. subtilis* PucM) [PMID: 16782815; 21795808].

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Step 3 (obligatory, conserved enzyme). OHCU is likewise unstable and would also decay to racemic allantoin. OHCU decarboxylase (UraD/PucL) decarboxylates OHCU and, via a stereospecific protonation step, produces **(S)-allantoin**. It is a **cofactor-independent**

homodimer with a novel all- α fold; an invariant histidine drives proton transfer from the C4 hydroxyl to C5 at the re-face of OHCU to set the (S) stereochemistry [PMID: 17567580;

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3.2 Obligatory vs. conditional vs. accessory steps

- **Obligatory:** All three steps are obligatory to reach (S)-allantoin efficiently. Without steps 2 and 3, the pathway still produces allantoin — but slowly, racemically, and via uncontrolled non-enzymatic decay [PMID: 16462750].
- **Conditional / variable:** The *identity* of the step-1 oxidase is conditional on lineage (Uox vs. PuuD vs. HpxO). Whether steps 2 and 3 are separate or fused is also lineage-conditional.
- **Accessory (outside the module):** Allantoin racemase, allantoinase, and further ureide enzymes are downstream accessories that determine the fate of allantoin but are not part of the three-reaction module.

4. Major Molecular Players and Active Assemblies

4.1 PuuD — the membrane cytochrome c urate oxidase (Finding F001)

PuuD is the defining and least-understood component. It is an **integral-membrane protein** whose transmembrane region (8 helices) corresponds to **DUF989 — a family with no similarity to known proteins** — carrying a **C-terminal cytochrome c** domain [PMID: 26349049]. Its discovery is itself notable: Doniselli et al. identified *puuD* not by homology but by combining gene-network analysis with association-rule learning across genomes, then confirmed function genetically. The gene-deletion/complementation experiment in *Agrobacterium fabrum* is the linchpin: loss of *puuD* abolishes urate degradation, and heterologous cytosolic zebrafish Uox restores it. The catalase-assay comparison — a 4-fold rise in the Uox-complemented strain versus no rise in wild type — is direct mechanistic evidence that PuuD "proceeds through cytochrome c-mediated electron transfer" rather than releasing H₂O₂ [PMID: 26349049]. As the authors put it, "in contrast with the known soluble Uox, the identified gene (*puuD*) encodes a membrane protein with a C-terminal cytochrome c," and "the 8-helix transmembrane domain corresponds to DUF989, a family without similarity to known proteins." This makes PuuD mechanistically distinct from every other known urate oxidase and, in the authors' phrase, completes "the catalytic repertoire of a therapeutic enzyme."

4.2 HIU hydrolase (PucM / UraH) — a transthyretin-related tetramer (Finding F003)

B. subtilis PucM is a 121-residue protein forming a **homotetramer** that "shows high sequence similarity to the functionally unrelated protein transthyretin (TTR)" and therefore "belongs to the TTR-related proteins (TRP) family" [PMID: 16782815]. The same fold and quaternary structure were confirmed in a second organism: *Klebsiella pneumoniae* HIU hydrolase (KpHIUH) is "a homotetrameric protein that, based on sequence and structural similarity, belongs to the transthyretin-related protein family," and its high-resolution structure plus active-site mutagenesis yielded a proposed catalytic mechanism [PMID: 21795808]. The cross-organism agreement (*Bacillus* and *Klebsiella*) makes the tetrameric TRP fold assignment robust. Conserved active-site residues include His-14, Arg-49, and His-105, with a C-terminal Tyr118-Arg119-Gly120-Ser121 motif positioned at the dimer interface.

4.3 OHCU decarboxylase (UraD / PucL) — a cofactor-independent novel fold (Finding F004)

OHCU decarboxylase was solved with bound (S)-allantoin as a **homodimeric helical protein representing a novel structural motif**, with an active site that "contains no cofactors, distinguishing this enzyme mechanistically from other cofactor-dependent decarboxylases" [PMID: 17567580]. The stereochemical logic is explicit: "the invariant histidine residue in the OHCU decarboxylase family plays an essential role in producing (S)-allantoin through a proton transfer from the hydroxyl group at C4 to C5 at the re-face of OHCU" [PMID: 17567580]. An independent 1.8 Å structure described the same novel all- α fold (ten α -helices in two domains per monomer) and identified His-67 and Glu-87 as essential — Glu-87 destabilizing the substrate ground state to facilitate decarboxylation, His-67 protonating to yield the (S)-enantiomer [PMID: 17428786]. Pharmacologically, the *K. pneumoniae* enzyme is competitively inhibited by allopurinol ($K_i \approx 30 \mu\text{M}$), the first reported OHCU decarboxylase inhibitor, which disrupts the necessary organization of the active site [PMID: 20826786].

4.4 Summary table of active assemblies

Component	Organism(s) characterized	Oligomeric state	Fold / family	Cofactor	Key catalytic residues
PuuD	<i>Agrobacterium fabrum</i>	Membrane protein	DUF989/COG3748 + C-terminal cyt c	Heme c (electron transfer)	Not yet resolved
HIU hydrolase (PucM/UraH)	<i>B. subtilis</i> , <i>K. pneumoniae</i>	Homotetramer	Transthyretin-related (TRP)	None	His-14, Arg-49, His-105; C-term YRGS
OHCU decarboxylase (UraD/PucL)	zebrafish, <i>K. pneumoniae</i>	Homodimer	Novel all- α	None	His-67, Glu-87 (invariant His)

5. Evolutionary and Cell-Biological Variation

5.1 Multiple non-homologous oxidases fill the step-1 slot (Finding F005)

The most striking variation is at step 1, where **three mechanistically distinct enzymes independently perform urate oxidation**:

1. **Soluble cofactor-independent Uox** — $O_2 \rightarrow H_2O_2$; cytosolic in bacteria, peroxisomal in animals.
2. **Membrane cytochrome c PuuD (DUF989)** — electron transfer, no H_2O_2 [PMID: 26349049].
3. **FAD-dependent HpxO** — in *K. pneumoniae*, "the proteins involved in the oxidation of ... uric acid (HpxO) did not display any similarity to other reported enzymes known to catalyze these reactions but instead are similar to oxygenases acting on aromatic compounds" [PMID: 18849434].

This is a textbook case of **independent (convergent) evolution onto a shared metabolic node**: the downstream chemistry (unstable HIU that must be captured) is identical, but the oxidase that generates it is drawn from three unrelated protein families. The PuuD-linked module is simply the version in which the front-end oxidase is the membrane cytochrome c enzyme.

5.2 Fusion of downstream steps (Finding F005)

The two downstream activities can be encoded by two proteins (bacteria such as *Bacillus* and *Klebsiella*) or **fused into a single bifunctional enzyme**. The *Arabidopsis thaliana* transthyretin-like (TTL) protein "is a tetrameric bifunctional enzyme, since it has 5-HIU hydrolase and OHCU decarboxylase activities" [PMID: 20167108]. This is a lineage-specific elaboration that couples the two trapping steps into one assembly, plausibly improving the handoff of the unstable OHCU intermediate.

5.3 Compartmentalization and cell-biological variation

The PuuD-linked step is, by definition, **membrane-associated** — a compartmental distinction from the soluble/peroxisomal Uox route. In teleost fish, HIU hydrolase exists as paralogous subfamilies distinguished by peroxisomal targeting: "HIUHase 1 (which has an N-terminal peroxisomal targeting signal-2 [PTS2]) and HIUHase 2 (which does not have an N-terminal PTS2)" [PMID: 23994290], indicating that even the conserved downstream step has undergone compartment-specific and gene-duplication-driven diversification in some lineages.

5.4 Loss in hominids

In hominids, including *Homo sapiens*, the pathway enzymes are generally not expressed — uricase (UOX) is pseudogenized, making uric acid the terminal product of purine catabolism and predisposing to hyperuricemia and gout [PMID: 17428786]. Notably, some downstream genes retain protein-coding transcripts: recombinant human OHCU decarboxylase is a well-folded α -helical protein highly similar to the zebrafish enzyme (52% sequence identity, RMSD < 0.8 Å) but with reduced stability and markedly lower catalytic activity — "reflecting the progressive relaxation of selective pressure following uricase loss in hominids" [PMID: 41956358].

6. Constraints, Dependencies, and Failure Modes

6.1 Obligatory ordering and the "unstable-intermediate trap" logic

The three steps are **strictly ordered by chemistry**: urate $\rightarrow$ HIU $\rightarrow$ OHCU $\rightarrow$ allantoin. Both HIU and OHCU are **unstable in solution and decay non-stereospecifically to allantoin** [PMID: 21616082]. The entire *raison d'être* of steps 2 and 3 is kinetic: the downstream enzymes trap these intermediates faster than they can decay, converting a slow, racemic, uncontrolled process into a fast, stereospecific one. This is why "urate oxidation produces racemic allantoin on a time scale of hours, whereas the full enzymatic complement produces dextrorotatory allantoin on a time scale of seconds" [PMID: 16462750].

6.2 Stereochemical constraint and downstream compatibility

The pathway must produce **(S)-allantoin** because the next enzyme in ureide catabolism, allantoinase, is specific for (S)-allantoin. Non-enzymatic racemization at physiological pH generates a mixture, so cells that rely on the non-stereospecific route additionally require an **allantoin racemase** to feed allantoinase [PMID: 21616082]. In the fully enzymatic module, the invariant histidine of OHCU decarboxylase enforces the (S) configuration directly [PMID: 17567580], removing the need for racemase correction.

6.3 Mechanistically mutually exclusive oxidase chemistries

A defining constraint is that the step-1 chemistries are mutually exclusive within a given enzyme: PuuD's cytochrome c electron transfer **does not** produce H_2O_2 , whereas soluble Uox **does**. The catalase-assay evidence rules out an H_2O_2 -producing mechanism for PuuD [PMID: 26349049]. This experimentally excludes the otherwise plausible assumption that all urate oxidases share the Uox mechanism, and it is the central mechanistic distinction that defines the PuuD-linked module as separate from the classical uricase route.

6.4 Compartment and pathway-boundary constraints

- PuuD is membrane-bound; the classical Uox route is soluble/peroxisomal — the routes are compartment-specific alternatives, not interchangeable within one location.
- The **anaerobic** gut route (urate $\rightarrow$ xanthine/SCFA) is oxygen-independent and product-divergent; it cannot substitute for the aerobic (S)-allantoin module and does not use HIU/OHCU intermediates [PMID: 37541197].

6.5 Failure modes

- **Loss of the oxidase** (e.g., hominid *UOX* pseudogenization) blocks the whole pathway → urate accumulation → gout [PMID: 17428786;

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- **Loss/decay of downstream enzymes** → intermediates escape as racemic allantoin, incompatible with (S)-specific allantoinase unless a racemase is present [PMID: 21616082].
- **Inhibition** — allopurinol competitively inhibits OHCU decarboxylase ($K_i \approx 30 \mu\text{M}$), disrupting active-site organization [PMID: 20826786].

7. Controversies, Open Questions, and Evolutionary Origin

7.1 What is strongly supported

- The **three-enzyme architecture** and the identity of the two intermediates (HIU, OHCU) are firmly established by phylogenetic genome comparison plus biochemistry [PMID: 16462750].
- The **fold assignments** of the two downstream enzymes are supported by multiple crystal structures across organisms [PMID: 16782815;

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- **PuuD's identity and electron-transfer mechanism** are supported by genetics, complementation, and the catalase differential [PMID: 26349049].

7.2 Evolutionary origin: HIU hydrolase is the ancestral transthyretin activity (Finding F008)

A particularly well-resolved evolutionary question concerns the transthyretin (TTR) / TRP family. **Ancestral-sequence reconstruction across 123 vertebrate sequences** shows that the pre-duplication ancestor "was unable to bind thyroxine and likely presented the modern HIUase reaction mechanism, while the substitutions after duplication prevented that activity

and were enough to provide stable thyroxine binding, as confirmed by calorimetry and x-ray diffraction" [PMID: 33956179]. Independent phylogenetic and functional work agrees: "Transthyretin (TTR) diverged from an ancestral 5-hydroxyisourate hydrolase (HIUHase) by gene duplication at some early stage of chordate evolution" [PMID: 25863347], with parallel evidence in trout showing HIUHase and TTR diverged from a common ancestral HIUHase with no functional complementation [PMID: 23994290]. **The mechanistic conclusion is that the enzymatic (HIUase) activity is ancestral and the famous thyroid-hormone-binding function of transthyretin is a derived, post-duplication elaboration.** For understanding the ancestral role of the family, the HIU-hydrolase members — not vertebrate hormone-binding TTR — are the best representatives.

The deepest plausible origin of the *module* is therefore the conserved downstream trap-and-decarboxylate machinery, whose components (TRP-family hydrolase; novel-fold cofactor-independent decarboxylase) are ancient and broadly distributed. The **oxidase step is the labile, repeatedly re-invented node** — solved independently by Uox, PuuD, and HpxO — and is best regarded as a later, lineage-specific "front end" bolted onto the conserved downstream core.

7.3 Where the literature is thin, indirect, or mixes organisms

- **PuuD structure and acceptor chemistry.** There is no atomic-resolution structure of PuuD, and its physiological electron acceptor and the coupling between the DUF989 transmembrane domain and the C-terminal cytochrome c remain unresolved. The mechanism is inferred largely from a single genetic/complementation study in one organism [PMID: 26349049].
- **Organism mixing.** Much of the mechanistic detail for steps 2 and 3 comes from *Bacillus*, *Klebsiella*, zebrafish, and *Arabidopsis*, while step 1 (PuuD) is characterized in *Agrobacterium*. Care is needed not to assume that a downstream enzyme's kinetics or targeting in one organism apply to the PuuD-bearing organism.
- **Naming ambiguity (Finding F006).** "PuuD" denotes two unrelated proteins. In *E. coli* K-12, PuuD is γ -glutamyl- γ -aminobutyrate hydrolase: "the puuD-deficient strain accumulated gamma-glutamyl-gamma-aminobutyrate (gamma-Glu-GABA) and could not grow on putrescine as a sole nitrogen source" [PMID: 16499623]. This has no connection to urate metabolism and is a persistent source of literature confusion.

7.4 Most important open questions

1. **What is the atomic structure of PuuD**, and how does its cytochrome c couple urate oxidation to the respiratory/electron-transport chain?
2. **What is the physiological electron acceptor** and the redox partner(s) of PuuD?
3. **How widespread is the PuuD (DUF989/COG3748) solution** across bacterial phyla relative to Uox and HpxO, and what ecological/physiological conditions select for a membrane, H₂O₂-free oxidase?
4. **Is there metabolic channeling** between PuuD and the downstream TRP hydrolase, given the instability of HIU?
5. **Why do some hominid downstream genes remain transcribed** (e.g., *URAD*) despite uricase loss — is there a cryptic or moonlighting function [PMID: 41956358]?

8. Limitations and Knowledge Gaps

This review is a literature synthesis of 18 primary papers and 8 confirmed findings; it did not involve new experimental data. The principal limitations are:

- **PuuD is under-characterized.** The central protein of the topic has no solved structure and only one detailed functional study. Its mechanism, while well-argued, rests on indirect (catalase-differential) evidence for the absence of H₂O₂ and on genetic complementation, rather than on direct spectroscopic or single-turnover kinetics.
- **Cross-organism extrapolation.** The three steps have been best characterized in *different* organisms. Statements about "the PuuD module" implicitly stitch together data from *Agrobacterium* (step 1), *Bacillus/Klebsiella* (step 2), and zebrafish/*Klebsiella* (step 3).
- **Distribution and regulation.** The genomic distribution, operon organization, and transcriptional regulation of PuuD-linked clusters are not well mapped; the regulatory data available (e.g., *hpx* clusters, [PMID: 18849434;

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pertain to HpxO-linked, not PuuD-linked, systems.

- **No quantitative meta-analysis** was possible; findings are qualitative/mechanistic rather than statistical effect-size comparisons.

9. Proposed Follow-up Experiments and Actions

1. **Determine the PuuD structure.** Pursue cryo-EM or crystallography of a PuuD homolog (e.g., from *Agrobacterium fabrum*) to resolve the DUF989 transmembrane domain and its interface with the C-terminal cytochrome c.
 2. **Identify PuuD's electron acceptor.** Reconstitute PuuD in proteoliposomes or membranes and test candidate acceptors (soluble cytochromes, quinones, terminal oxidases) with stopped-flow/spectroelectrochemistry; directly confirm the absence of H₂O₂ using amperometric peroxide detection during single turnover.
 3. **Comparative genomics of the oxidase node.** Systematically census DUF989/COG3748 (PuuD), Uox, and HpxO across sequenced bacterial genomes; correlate oxidase type with aerobiosis, habitat, and co-occurring downstream genes to test the "interchangeable front-end" model.
 4. **Test metabolic channeling.** Co-express PuuD with the downstream TRP hydrolase and assay whether HIU is handed off directly (reduced racemic-allantoin leakage) versus released to bulk solution.
 5. **Reconstitute the full PuuD-linked pathway** in vitro or in a heterologous host and measure the rate and stereochemical purity of (S)-allantoin production, benchmarking against the Uox-initiated pathway.
 6. **Probe hominid URAD residual function.** Given retained transcription and reduced-but-present folding of human OHCU decarboxylase [PMID: 41956358], test for any residual decarboxylase or moonlighting activity and its physiological relevance.
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10. Key References

- Ramazzina I, et al. *Completing the uric acid degradation pathway through phylogenetic comparison of whole genomes*. PMID: 16462750 — Defines the three-step relay; racemic-over-hours vs. (S)-in-seconds.
- Doniselli N, et al. *The identification of an integral membrane, cytochrome c urate oxidase completes the catalytic repertoire of a therapeutic enzyme*. PMID: 26349049 — Discovery and mechanism of PuuD (DUF989, cytochrome c, no H₂O₂).

- Jung DK, et al. *Structural and functional analysis of PucM, a hydrolase in the ureide pathway and a member of the transthyretin-related protein family*. PMID: 16782815 — HIU hydrolase is a TRP-family homotetramer (*B. subtilis*).
- French JB, Ealick SE. *Structural and kinetic insights into the mechanism of 5-hydroxyisourate hydrolase from Klebsiella pneumoniae*. PMID: 21795808 — Confirms tetrameric TRP fold in a second organism.
- Kim K, et al. *Structural and functional basis for (S)-allantoin formation in the ureide pathway*. PMID: 17567580 — Cofactor-independent novel-fold OHCU decarboxylase; invariant His sets (S) stereochemistry.
- Cendron L, et al. *The structure of 2-oxo-4-hydroxy-4-carboxy-5-ureidoimidazoline decarboxylase provides insights into the mechanism of uric acid degradation*. PMID: 17428786 — 1.8 Å novel all- α fold; His-67 and Glu-87 essential.
- French JB, Ealick SE. *Structural and mechanistic studies on Klebsiella pneumoniae OHCU decarboxylase*. PMID: 20826786 — Allopurinol competitive inhibitor ($K_i \approx 30 \mu\text{M}$).
- de la Riva L, et al. *The hpx genetic system for hypoxanthine assimilation as a nitrogen source in Klebsiella pneumoniae*. PMID: 18849434 — FAD/oxygenase-type urate oxidase HpxO as an alternative step-1 enzyme.
- Lamberto I, et al. *Functional characterization of Arabidopsis thaliana transthyretin-like protein*. PMID: 20167108 — Bifunctional fusion of steps 2 and 3.
- Kurihara S, et al. *Gamma-glutamyl-gamma-aminobutyrate hydrolase in the putrescine utilization pathway of Escherichia coli K-12*. PMID: 16499623 — The unrelated *E. coli* PuuD; naming ambiguity.
- Liu Y, et al. *A widely distributed gene cluster compensates for uricase loss in hominids*. PMID: 37541197 — Distinct anaerobic urate $\rightarrow$ xanthine/SCFA route.
- Carrijo de Oliveira L, et al. *Reenacting the Birth of a Function: Functional Divergence of HIUases and Transthyretins as Inferred by Evolutionary and Biophysical Studies*. PMID: 33956179 — Ancestral-reconstruction evidence that HIUase is the ancestral TTR-family activity.
- *Characterization of little skate (Leucoraja erinacea) recombinant transthyretin*. PMID: 25863347 — TTR diverged from ancestral HIUHase in early chordates.
- *Characterization of Oncorhynchus mykiss 5-hydroxyisourate hydrolase/transthyretin superfamily*. PMID: 23994290 — HIUHase paralog subfamilies (PTS2 vs. non-PTS2).
- *Characterization of the structure and function of Klebsiella pneumoniae allantoin racemase*. PMID: 21616082 — Instability of HIU/OHCU; need for (S)-specific product.

- *Structural conservation, functional decline: The fate of human OHCU decarboxylase.* [PMID: 41956358](#) — Human URAD retains folded but low-activity enzyme post-uricase loss.
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Prepared as a commissioned review synthesis. Confidence is highest for the three-enzyme architecture, the fold assignments of the two downstream enzymes, and the ancestral HIUase status within the transthyretin family; it is lowest for the atomic mechanism and structure of PuuD itself, which remains the key experimental frontier.

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